

Controlling the Accessible Lie Dimension Ratio in Neural Network Training

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Abstract

Neural network training evolves in a high-dimensional parameter space under sample gradient fields. A natural but insufficiently characterized property is the ratio between the dimension generated by these fields and the full parameter dimension. We define and study this quantity through the accessible Lie dimension. Our theoretical analysis and numerical measurements show that, for analytic architectures, almost every parameter point attains the architecture-level maximum, and that commonly used operators and architectures exhibit a high accessible Lie-dimension ratio. We develop two constructive routes for lowering this ratio. The first route uses symmetry to constrain training to a lower-dimensional invariant subset, yielding Pairing initialization under permutation symmetry, which forces exchangeable parameters into a symmetry-fixed subspace that remains strictly preserved throughout optimization. The second lowers the architecture-level maximum itself. For this route, we establish a general first-integral-based design principle for parameterizations whose induced constraints are preserved by the generated Lie algebra, and instantiate it in the Householder Orthogonal Vision Network (HOVNET). Both constructions achieve competitive empirical performance, indicating substantial potential for studying and designing neural networks through accessible Lie dimension.

1 Introduction

Modern neural networks are highly overparameterized, yet their training dynamics are far from arbitrary. Studies of intrinsic dimension and low-dimensional optimization trajectories show that successful training can be confined to a small part of parameter space (Li et al., 2018; Gur-Ari et al., 2018; Li et al., 2023). A complementary line of work studies implicit bias, conservation laws, invariant structures, and parameter symmetries to identify properties selected or preserved by gradient-based optimization (Marcotte et al., 2023; Godfrey et al., 2022). These studies reveal substantial structure in neural network training, but their principal objects are realized trajectories, preserved quantities, or parameter equivalence. They do not directly describe how many local parameter directions a given architecture allows its training dynamics to generate.

This work asks a direct question: at a parameter point, how many independent directions can be generated by sample gradients and their local interactions? In neural network training, each training sample induces a gradient vector field in parameter space. The span of the current gradients describes the directions available to one infinitesimal update. Successive updates generated by different samples or mini-batches, however, generally do not commute and can produce new local directions beyond the span of the current gradients. The closure of the sample gradient fields under these interactions is described by their generated Lie algebra. We call the dimension of its pointwise span the accessible Lie dimension. This quantity measures the number of local parameter directions that repeated gradient updates can generate.

For each fixed analytic architecture, there is a largest accessible Lie dimension permitted by its parameterization. We prove that the parameter points at which the architecture-level maximum is attained form an open, full-measure subset of parameter space. An initialization drawn from a continuous distribution therefore belongs to the maximal-dimension stratum with probability one. Under update rules that preserve

absolute continuity, this maximality also holds almost surely along any finite sequence of training iterates; our numerical measurements are consistent with this prediction. The tested multilayer perceptron and convolutional operators attain a sampled accessible Lie dimension equal to their parameter dimension, while the tested compact Vision Transformer yields a ratio nearly equal to 1. These results suggest that common operators exhibit an almost full-dimensional accessible structure, whereas lower ratios may indicate latent architecture-specific constraint equations.

Representation learning is commonly motivated by the premise that useful factors of variation have an intrinsic dimension substantially below the ambient dimension of the observations (Bengio et al., 2013; Tenenbaum et al., 2000; Fefferman et al., 2016). This motivates us to ask whether effective neural networks can be constructed with an accessible Lie-dimension ratio substantially below one.

We develop two constructive routes to a lower accessible Lie-dimension ratio. The first route keeps the architecture fixed and confines training to a lower-dimensional invariant subset that is preserved throughout optimization. We realize this by exploiting symmetry-fixed subspaces: specializing the symmetry action to neuron or channel permutations yields Pairing initialization, a mechanism that ties exchangeable parameter blocks at initialization and inherently maintains this lower-dimensional trajectory under standard gradient updates. The second route reduces the architecture-level maximum by introducing architectural constraints. We instantiate this strategy with a scale-invariant Householder orthogonal operator, which forms the basis of HOVNET.

Experiments show that several Pairing configurations outperform their baselines, while Householder-based models remain competitive on CIFAR-100 and Tiny ImageNet and can replace a substantial part of attention-based mixing.

This work makes three contributions. First, it defines accessible Lie dimension and establishes its generic maximality over parameter space. Second, it develops two principled mechanisms for lowering its ratio through symmetry-fixed subspaces and operator-level first integrals. Third, empirical evaluations provide evidence for the practical viability and broader design potential of accessible-dimension control.

2 Related Work

Training dimension and training geometry. While previous studies on intrinsic dimension, mean-field limits, and conservation laws primarily observe realized optimization trajectories or limiting behaviors (Li et al., 2018; Gur-Ari et al., 2018; Jacot et al., 2018; Marcotte et al., 2023; Zhao et al., 2026), our approach proactively quantifies and constrains the generated gradient fields. Drawing upon the Hermann–Nagano theorem for analytic vector fields (Hermann, 1963; Nagano, 1966; Warner, 1983), we provide a mathematically rigorous framework to measure structural capacity.

Symmetry and structured parameterization. Permutation symmetry produces equivalent parameter descriptions in neural networks (Godfrey et al., 2022). Orthogonal and unitary parameterizations have been used to control recurrent dynamics and learned operators (Hyland & Rätsch, 2017; Saxe et al., 2014; Householder, 1958). Low-rank adaptation reduces trainable parameters through matrix factorization (Hu et al., 2022). These methods impose different restrictions on parameter space, optimizer updates, or operator structure. Our perspective places these restrictions in a common geometric framework through the Lie algebra generated by training fields.

Compact vision architectures. ViT and DeiT established patch-based image Transformers and data-efficient training (Dosovitskiy et al., 2021; Touvron et al., 2021a). PVT, NesT, CvT, and CCT introduced hierarchical token processing and convolutional inductive biases (Wang et al., 2021; Zhang et al., 2022; Wu et al., 2021; Hassani et al., 2021a). Recent works further adapt Transformers for small datasets or replace attention entirely for computational efficiency (Lu et al., 2022; Zhang & Zhang, 2025; Nguyen et al., 2025; Touvron et al., 2021b; Zhang et al., 2025; Mehta & Rastegari, 2022). In contrast to these architectural modifications, we view operator design fundamentally as a mechanism for controlling the geometry of accessible training directions.

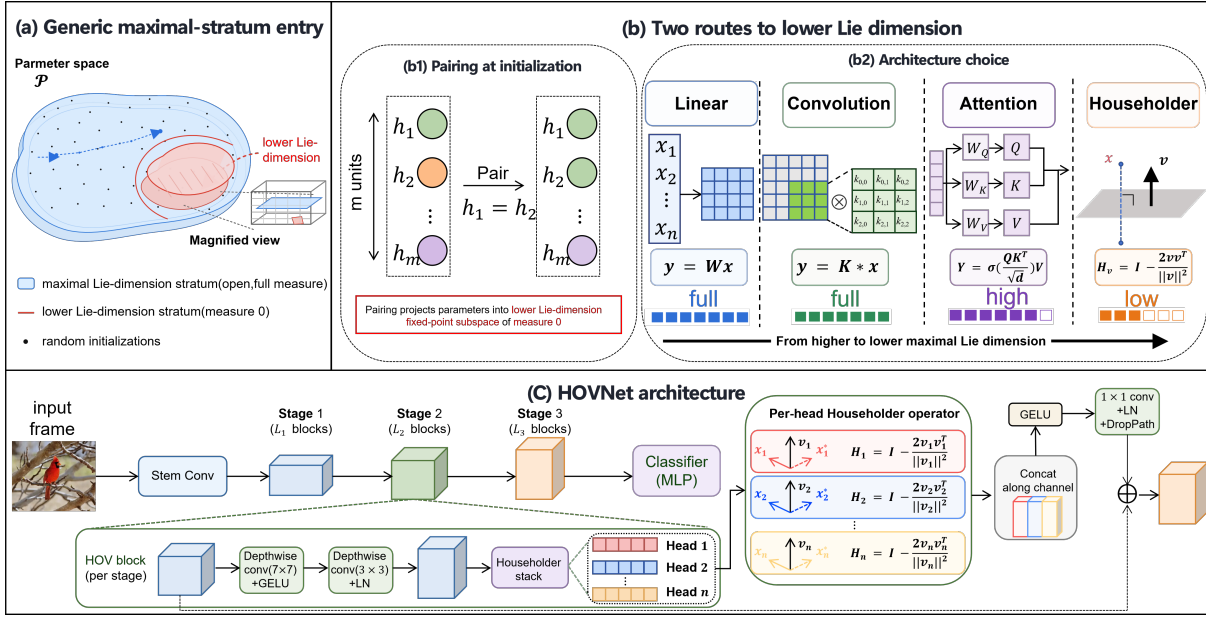


Figure 1: Overview of the theory and the proposed operator. Panel (a) shows the full-measure maximal-ratio stratum and the measure-zero lower-ratio strata. Panel (b1) shows Pairing at initialization. Panel (b2) illustrates operator-induced parameter constraints, whose network-level effect depends on the surrounding architecture. Panel (c) shows the HOVNET architecture and its multihead Householder block.

3 Accessible Lie Dimension

We state the main theoretical results in this section and provide all proofs in the supplementary material.

Let $f_\theta : \mathcal{X} \rightarrow \mathbb{R}^c$ be a model with parameter vector $\theta \in \Theta \subseteq \mathbb{R}^M$.

Definition 1 (Accessible Lie dimension). For $x \in \mathcal{X}$ and $a \in \mathbb{R}^c$, let $X_{x,a}(\theta) = \nabla_\theta \langle a, f_\theta(x) \rangle$ and $\mathcal{W} = \{X_{x,a} : x \in \mathcal{X}, a \in \mathbb{R}^c\}$. The accessible Lie dimension is $d_{\text{Lie}}(\theta) = \dim \mathcal{D}(\theta)$, and the ratio is $\rho_{\text{Lie}}(\theta) = d_{\text{Lie}}(\theta)/M$, where $\mathcal{D}(\theta) = \text{span}\{Y(\theta) : Y \in \text{Lie}(\mathcal{W})\}$ is the accessible distribution.

3.1 Generic Maximality

For a fixed architecture, the parameter dimension M is constant, so maximizing d_{Lie} is equivalent to maximizing ρ_{Lie} . The accessible Lie dimension is generically constant on an open, full-measure stratum of parameter space, as illustrated in Figure 1(a). The precise statement is given below.

Theorem 1 (Generic maximality). Let $\Theta \subseteq \mathbb{R}^M$ be connected open, and assume that every vector field in $\mathcal{W}$ is real analytic on Θ . Define

$$R = \max_{\theta \in \Theta} d_{\text{Lie}}(\theta), \quad \Omega_R = \{\theta \in \Theta : d_{\text{Lie}}(\theta) = R\}.$$

Then Ω_R is open and $\Theta \setminus \Omega_R$ has Lebesgue measure zero. Moreover, if μ is a probability measure on Θ that is absolutely continuous with respect to Lebesgue measure and $\theta_0 \sim \mu$, then

$$\Pr_{\theta_0 \sim \mu}(\theta_0 \in \Omega_R) = \mu(\Omega_R) = 1.$$

Remark 1. The analyticity assumption in Theorem 1 is satisfied by a broad class of commonly used architectures whose constituent operations are real analytic in the parameters. The same conclusion holds when $\mathcal{W}$ is replaced by any fixed finite family of analytic gradient fields. Lie brackets account for the local noncommutativity of the corresponding sample or mini-batch updates.

Since $\Theta \setminus \Omega_R$ is a null set, any finite collection of iterates $\theta_0, \dots, \theta_T$ whose marginal distributions are absolutely continuous satisfies

$$\Pr \left(\bigcap_{t=0}^T \{\theta_t \in \Omega_R\} \right) = 1.$$

Thus, the accessible Lie dimension remains maximal almost surely throughout any finite training horizon, provided that the update rule preserves absolute continuity at every finite time.

3.2 Architecture-Level Reduction via First Integrals

For a smooth map $C : \Theta \rightarrow \mathbb{R}^k$, let $dC_\theta : \mathbb{R}^M \rightarrow \mathbb{R}^k$ denote its differential at θ . We call C a regular common first integral of $\mathcal{W}$ if $\text{rank } dC_\theta = k$ and $dC_\theta(X_{x,a}(\theta)) = 0$, $\forall \theta \in \Theta$, $x \in \mathcal{X}$, and $a \in \mathbb{R}^c$.

Proposition 1 (First-integral restriction). *If C is a regular common first integral of $\mathcal{W}$, then $dC_\theta(Y(\theta)) = 0$, $\forall \theta \in \Theta$ and $Y \in \text{Lie}(\mathcal{W})$. Consequently, $\mathcal{D}(\theta) \subseteq \ker dC_\theta$, $d_{\text{Lie}}(\theta) \leq M - k$, and $\rho_{\text{Lie}}(\theta) \leq 1 - k/M$.*

Example 1 (Bilinear factorization). For $\phi(Q, K) = QK^\top$, with $Q \in \mathbb{R}^{m \times r}$ and $K \in \mathbb{R}^{n \times r}$, let $C(Q, K) = \text{vech}(Q^\top Q - K^\top K)$, where vech vectorizes the independent entries of a symmetric matrix. Then C is a common first integral for all gradient fields induced by scalar probes depending on Q and K only through $\phi(Q, K)$. Wherever $\text{rank } dC_{(Q,K)} = k$, Proposition 1 yields $d_{\text{Lie}} \leq M - k$.

This operator-level route lowers the architecture-level maximum. We next consider the complementary fixed-architecture route.

3.3 Fixed-Architecture Reduction via Symmetry

Let $\rho : \Gamma \rightarrow O(M)$ be an orthogonal representation satisfying $\rho(\gamma)\Theta = \Theta$, and define

$$F_\Gamma := \{v \in \mathbb{R}^M : \rho(\gamma)v = v, \forall \gamma \in \Gamma\}, \quad \Theta^\Gamma := \Theta \cap F_\Gamma.$$

We say that the model is Γ -invariant if $f_{\rho(\gamma)\theta} = f_\theta$ on $\mathcal{X}$ for every $\gamma \in \Gamma$ and $\theta \in \Theta$.

Proposition 2 (Symmetry restriction and preservation). *Suppose that the model is Γ -invariant. Then $\mathcal{D}(\theta) \subseteq F_\Gamma$, $d_{\text{Lie}}(\theta) \leq \dim F_\Gamma$ for every $\theta \in \Theta^\Gamma$. Consequently, $\rho_{\text{Lie}}(\theta) \leq \frac{\dim F_\Gamma}{M} < 1$ on Θ^Γ whenever $F_\Gamma \subsetneq \mathbb{R}^M$.*

Moreover, if $\theta_0 \in \Theta^\Gamma$ and each update $\Delta_t \in F_\Gamma$, then every iterate of $\theta_{t+1} = \theta_t + \Delta_t$ remains in Θ^Γ . Hence, the same dimension bounds hold throughout training.

Remark 2. The update condition in Proposition 2 is automatically satisfied by standard gradient-based optimizers under the corresponding symmetry assumptions.

Example 2 (Permutation symmetry). Permutations of functionally equivalent neurons, channels, or attention heads satisfy the invariance condition in Proposition 2. Standard optimizers, including SGD, Adam, and AdamW, are equivariant under such permutations when the associated optimizer states and hyperparameters are transformed consistently.

The preceding results give two abstract but constructive routes to a lower accessible Lie-dimension ratio. The next two sections develop their practical implementations. Section 4 first introduces a sampled Lie algebra rank condition (LARC) procedure for estimating accessible dimension and then realizes the permutation construction in Example 2 as Pairing initialization. Section 5 specializes Proposition 1 to Householder reflections, whose scale invariance supplies an explicit common first integral.

4 LARC and Pairing Initialization

This section introduces a sampled LARC estimator for accessible dimension and then realizes the fixed-architecture route through Pairing initialization.

Algorithm 1 Sampled LARC with first-order Lie brackets**Require:** Model f_θ , probes $\mathcal{P}_0$, bracket pairs $\mathcal{P}_1$, tolerances $\tau_{\text{abs}}, \tau_{\text{rel}} > 0$ **Ensure:** Sampled rank r_{samp}

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1: Initialize an empty orthonormal basis  $Q$ 
2: for each probe  $i \in \mathcal{P}_0$  do
3:   Evaluate  $b \leftarrow X_i(\theta) = \nabla_\theta \langle a_i, f_\theta(x_i) \rangle$ 
4:   Orthogonalize  $b$  twice against  $Q$ , giving residual  $b_\perp$ 
5:   if  $\|b_\perp\|_2 > \max(\tau_{\text{abs}}, \tau_{\text{rel}}\|b\|_2)$  then
6:     Normalize and append  $b_\perp$  to  $Q$ 
7:   end if
8: end for
9: for each  $(i, j) \in \mathcal{P}_1$  do
10:  Compute  $b \leftarrow [X_i, X_j](\theta)$  by Hessian–vector products
11:  Apply the same orthogonalization and acceptance rule
12: end for
13: return  $r_{\text{samp}} = \dim \text{span}(Q)$ 

```

Sampled LARC estimation. Exact evaluation of $\mathcal{D}(\theta)$ requires all iterated Lie brackets and is generally intractable. We therefore select a finite set of probes (x_i, a_i) , where x_i is an input example and $a_i \in \mathbb{R}^c$ is an output direction used to scalarize the model output. Each probe defines $g_i(\theta) = \langle a_i, f_\theta(x_i) \rangle$ and the sample-gradient field $X_i(\theta) = \nabla_\theta g_i(\theta)$. The set $\mathcal{P}_0$ specifies which gradient fields are evaluated, while $\mathcal{P}_1$ specifies selected pairs whose first-order brackets are added. Using $[X_i, X_j] = DX_j X_i - DX_i X_j$, we compute

$$[X_i, X_j](\theta) = \nabla_\theta^2 g_j(\theta) X_i(\theta) - \nabla_\theta^2 g_i(\theta) X_j(\theta)$$

by Hessian–vector products, without forming dense Hessians. Algorithm 1 incrementally keeps only numerically independent directions. The returned rank r_{samp} is therefore a probe- and tolerance-dependent lower bound on $d_{\text{Lie}}(\theta)$, and is exact when the sampled vectors span $\mathcal{D}(\theta)$.

Pairing initialization. We realize a permutation-group special case of Proposition 2 by assigning identical initial values to functionally exchangeable parameter blocks, thereby placing the model in a permutation-fixed subspace. Pairing is applied only after the complete unpaired model has been initialized, so the architecture, raw parameter count, and base initialization rule remain unchanged. The pairs are not chosen arbitrarily: only units related by a valid parameter permutation are paired, and all associated incoming, bias, and outgoing parameters must be copied together. This preserves the intended symmetry while avoiding partial pairings that could alter the model structure or initialization statistics in an uncontrolled way.

For example, consider two consecutive MLP layers $z = \phi(Wx + b)$ and $y = Uz + c$. Pairing hidden neurons i and j sets

$$W_{j\cdot} = W_{i\cdot}, \quad b_j = b_i, \quad U_{\cdot j} = U_{\cdot i}.$$

The paired neurons then have identical incoming parameters, activations, and outgoing parameter blocks; these are exactly the fixed-point conditions for swapping the two neurons. Hence the initialization lies in the symmetry-fixed subspace illustrated in Figure 1(b1), and standard symmetry-equivariant gradient updates preserve this relation throughout training. Because some valid pairings can impose an overly strong restriction, we design the pairing pattern separately for each architecture and initialization family, aiming to approach the target ratio while preserving the original model and initialization setting as closely as possible. Exact assignments are provided in the supplementary material.

5 Householder Orthogonal Vision Network

Proposition 1 bounds the architecture-level maximum whenever the parameterization induces independent common first integrals. The Householder reflector provides such a constraint through scale invariance. To evaluate this operator-level reduction principle in a practical vision setting, we use Householder reflectors

to construct the Householder Orthogonal Vision Network (HOVNET). Figure 1(c) presents the resulting network, while Figure 1(b2) illustrates the local parameterization constraint that motivates its construction.

5.1 Householder Orthogonal Operator

Let $v \in \mathbb{R}^d \setminus \{0\}$ be a trainable parameter vector and $z \in \mathbb{R}^d$ an input feature vector. The parameterized Householder operator maps z to the output

$$H_v(z) = H(v)z = z - 2v \frac{v^\top z}{v^\top v}, \quad H(v) = I_d - 2 \frac{vv^\top}{v^\top v}, \quad (1)$$

where I_d is the $d \times d$ identity matrix. For fixed v , $H(v)$ is an orthogonal linear map that reflects z across the hyperplane orthogonal to v ; hence

$$H(v)^\top H(v) = I_d, \quad \|H_v(z)\|_2 = \|z\|_2.$$

Given r trainable vectors $V = (v_1, \dots, v_r)$, the corresponding reflector stack is

$$Q_V(z) = Q(V)z, \quad Q(V) = H(v_r) \cdots H(v_1). \quad (2)$$

It stores rd parameters and requires $O(rd)$ operations per input vector. By contrast, an unconstrained dense $d \times d$ transformation requires d^2 parameters and $O(d^2)$ operations.

The parameterization is invariant under nonzero rescaling:

$$H(\alpha v) = H(v), \quad \alpha \neq 0. \quad (3)$$

Differentiating along the radial direction gives $DH(v)[v] = 0$. Consequently, every sample-gradient field induced by a scalar probe whose dependence on v is solely through $H(v)$ is orthogonal to v . The function

$$C(v) = \frac{1}{2} \|v\|_2^2$$

is therefore a common first integral of all sample gradient fields associated with this parameter block. By Proposition 1, every generated Lie bracket also annihilates C , so the accessible distribution is tangent to the level set of $\|v\|_2$.

The operator-level analysis identifies the structural mechanism by which the Householder parameterization constrains accessible directions. By Proposition 1, this explicit common first integral yields a strict theoretical upper bound formula for the architecture-level accessible Lie dimension, which we derive and detail in the supplementary material.

5.2 HOVNet Architecture

As one concrete realization of the Householder operator in a vision architecture, we construct HOVNET to evaluate the Proposition 1 and the proposed design principle. Because the Householder stack requires only $O(rC)$ parameters, it is insufficiently expressive to form a standalone building block. Its design therefore assigns spatial interaction primarily to depthwise convolutions and channel mixing primarily to Householder transformations, while pointwise 1×1 readouts provide necessary cross-head fusion.

As shown in Figure 1(c), the resulting network has a three-stage hierarchical architecture. A convolutional stem—comprising two stacked 3×3 convolutions with BatchNorm and GELU—produces the initial feature map, while transition layers reduce spatial resolution and increase channel width between stages. Stage $s \in \{1, 2, 3\}$ contains L_s blocks of width C_s . The configuration used in our experiments is

$$(C_1, C_2, C_3) = (192, 328, 448), \quad (4)$$

$$(L_1, L_2, L_3) = (4, 6, 10). \quad (5)$$

Unless specified otherwise, we set the number of channel heads $h = 8$ and the number of reflections per stack $r = 6$. Global average pooling and a linear classifier follow the final stage.

Consider one block with input $X \in \mathbb{R}^{B \times C \times H \times W}$, where B , C , H , and W denote the batch size, channel width, height, and width, respectively. Spatial mixing is performed by

$$S(X) = \text{DWConv}_{3 \times 3}(\text{GELU}(\text{DWConv}_{7 \times 7}(X))). \quad (6)$$

Both convolutions are depthwise and therefore exchange information across spatial positions without mixing channels. We then apply channel-wise layer normalization: $Z = \text{LN}_C(S(X))$.

Let h be the number of channel heads and assume that h divides C . Set $d = C/h$ and split Z along the channel dimension:

$$Z = \text{Concat}_{j=1}^h Z^{(j)}, \quad Z^{(j)} \in \mathbb{R}^{B \times d \times H \times W}.$$

For head j , let $v_{j,k} \in \mathbb{R}^d \setminus \{0\}$ be the parameter of its k -th reflector, where $k = 1, \dots, r$. The reflector stack acts independently at every batch and spatial index:

$$U_0^{(j)} = Z^{(j)}, \quad (7)$$

$$U_{k,b,:,p,q}^{(j)} = H(v_{j,k})U_{k-1,b,:,p,q}^{(j)}, \quad k = 1, \dots, r. \quad (8)$$

Here, $b \in \{1, \dots, B\}$, $p \in \{1, \dots, H\}$, and $q \in \{1, \dots, W\}$. Across all heads, the Householder stack uses $hrd = rC$ parameters and requires $O(rC)$ operations per spatial position.

The transformed heads are concatenated and fused by a learned pointwise 1×1 readout R :

$$V = \text{Concat}_{j=1}^h U_r^{(j)}, \quad (9)$$

$$P = R(\text{GELU}(V)), \quad (10)$$

$$\mathcal{B}(X) = X + \text{DropPath}(\gamma \odot \text{LN}_C(P)), \quad (11)$$

where $\gamma \in \mathbb{R}^C$ is a learnable channelwise scale initialized to 10^{-4} , and $\odot$ denotes channelwise multiplication. Stochastic depth (DropPath) is applied with a maximum rate of 0.20 linearly scaled across stages. This design systematically decouples spatial mixing (depthwise convolutions), within-head orthogonal channel mixing (Householder stacks), and cross-head fusion (1×1 readout). Compared to standard unconstrained convolutional networks, embedding the Householder operator effectively lowers the accessible Lie-dimension ratio across the architecture.

6 Experiments

We use EMNIST(Cohen et al., 2017), CIFAR-10, CIFAR-100 (Krizhevsky, 2009), and Tiny ImageNet (Le & Yang, 2015). For the reproduced classification comparisons, models within the same dataset and input-resolution setting use the same training duration, effective batch size, augmentation pipeline, learning rate schedule, and evaluation protocol as HOVNET. For each architecture, we retain only the optimizer family. MACs denote one multiply-accumulate operation for one forward pass with batch size one.

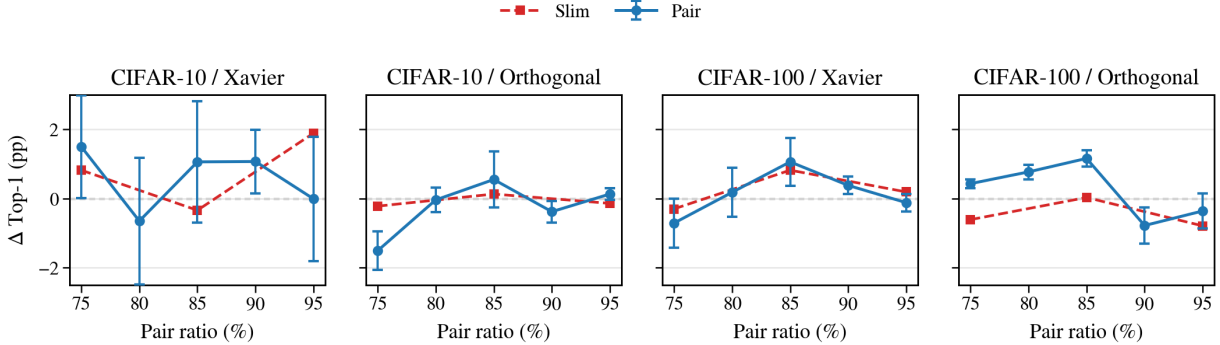
6.1 Accessible Lie Dimension During Training

We evaluate four common base architectures on the binary EMNIST task: two-layer MLPs, a compact CNN composed of repeated 3×3 convolution-SiLU blocks with spatial downsampling and global pooling, a compact Vision Transformer (ViT) built from pre-normalized multihead self-attention and feed-forward blocks, and a classifier assembled from the HOVNET block. All models are trained for 100 epochs, and the sampled first-order LARC is evaluated at initialization and at the final checkpoint using the same fixed probe plan. The Pairing variants adopt the same ViT architecture and differ only in their initialization.

Table 1 summarizes the results. The MLP and CNN baselines are full dimensional, whereas the unpaired ViT initially attains a sampled accessible ratio of approximately 94.4%. The complete HOVNET-block classifier

Table 1: Sampled LARC dimensions at initialization and after 100 training epochs.

Model	M	Init. rank	Final rank	Δ
MLP 64,64, GELU	12,481	12,481	12,481	0
MLP 64,64, tanh	12,481	12,481	12,481	0
MLP 64,128, GELU	16,705	16,705	16,705	0
MLP 64,128, tanh	16,705	16,705	16,705	0
CNN, SiLU	7,537	7,537	7,537	0
ViT	14,954	14,117	14,236	+119
HOVNET blocks	15,538	14,299	14,318	+19
ViT Pair 75%	14,954	10,523	10,638	+115
ViT Pair 95%	14,954	13,456	13,566	+110

Figure 2: Pairing results using CVT-7/4 on CIFAR-10 and CIFAR-100. Values are test-accuracy changes relative to the corresponding unpaired baseline; curves show mean $\pm$ standard deviation over three seeds.

attains a lower sampled ratio of approximately 92.0%. For Pairing, the predicted maximal dimension is obtained by scaling the unpaired ViT maximum according to the corresponding symmetry-fixed parameter ratio, and the measured ranks agree with these predictions within the expected estimation error.

The changes between initialization and the final checkpoint are relatively small compared to the corresponding parameter dimensions. While the ViT models show a slight increase in accessible dimension (e.g., $\Delta = +119$ for the base ViT) and HOVNET shows a minimal change ($\Delta = +19$), these deviations remain at the sub-percent level. Since sampled LARC is a finite-probe numerical estimator, such small deviations are treated as acceptable numerical error and are consistent with the theoretical preservation of the accessible ratio during training.

6.2 Pairing Experiments

We evaluate Pairing using the CVT-7/4 model from Compact Transformers (Hassani et al., 2021b), which is designed for small-scale image classification and provides a standard Transformer baseline on CIFAR-10 and CIFAR-100. In all runs, we retain the original model and training configuration and modify only the initialization by applying Pairing after Xavier or orthogonal initialization. On CIFAR-10, the unpaired Xavier and orthogonal baselines achieve test accuracies of 80.95% and 88.96%, respectively; the corresponding CIFAR-100 accuracies are 64.62% and 68.60%.

Figure 2 reports accuracy changes relative to these baselines over five pairing ratios and three random seeds. Pair denotes the symmetry-fixed model, while Slim refers to an unpaired, width-reduced model whose effective parameter count matches that of the Pair model.

Table 2: CIFAR-100 comparison under 2400-epoch training on 32×32 images. Reported metrics reflect the best accuracy between raw and EMA checkpoints.

Method	Params (M)	MACs (G)	Top-1 (%)
<i>Convolutional networks</i>			
ResNet-20	0.2783	0.0408	69.09
ResNet-56	0.8616	0.1258	78.04
WRN-28-4	5.8722	0.8456	80.70
ResNet-32 $\times$ 4	7.4510	1.1010	81.08
ConvNeXt-Tiny-CIFAR	27.8950	1.4530	81.78
<i>Vision Transformers</i>			
DeiT-Tiny/2	5.4101	1.6690	75.41
ViT-T/2	5.4100	1.6690	83.60
ViT-T/1	5.5560	10.2830	85.10
<i>Householder model and matched ablations</i>			
HOVNet (Ours)	5.5930	1.1300	84.10
HOVNET-MHSA	6.5997	1.4796	82.16
HOVNET-HeadLinear	5.8997	1.1666	82.95

Performance varies non-monotonically with the effective parameter ratio, peaking near 85%. This non-monotonic behavior illuminates a fundamental trade-off between optimization stability and expressive capacity. By constraining the accessible Lie dimension through symmetry (Pairing), the training trajectory is strictly confined to a lower-dimensional invariant subspace. In the moderate pairing regime, this structural restriction actively suppresses noisy or redundant gradient directions, thereby serving as an implicit regularizer that enhances generalization (improving upon the best baseline by up to 1.50 and 1.16 percentage points on CIFAR-10 and CIFAR-100, respectively). However, excessive parameter tying over-constrains the representational volume, starving the model of the capacity necessary to capture complex data distributions. Furthermore, the distinct superiority of the Pair model over the naively width-reduced Slim model demonstrates that optimizing within a structurally constrained, high-dimensional parameter space provides fundamentally different—and superior—gradient dynamics compared to simply initializing a smaller, unconstrained architecture.

6.3 CIFAR-100 Classification

Table 2 compares HOVNET with standard ResNet and Wide ResNet baselines (He et al., 2016; Zagoruyko & Komodakis, 2016), together with more recent ConvNeXt, DeiT, and ViT architectures (Liu et al., 2022; Touvron et al., 2021a; Zhang & Zhang, 2025). All models are trained on the standard 32×32 CIFAR-100 data for 2400 epochs. HOVNET uses AdamW with an initial learning rate of 3×10^{-3} , weight decay of 0.1, an EMA decay rate of 0.9999, 20 warmup epochs, cosine decay, and an effective batch size of 1024.

Data processing. All reproduced models use random cropping, horizontal flipping, RandAugment, Random Erasing, Mixup, and CutMix with the same hyperparameters. Mixup and CutMix are disabled during the final 20% of training. Reported results evaluate the higher Top-1 accuracy between raw and EMA checkpoints.

HOVNET achieves 84.10% top-1 accuracy, yielding a highly favorable accuracy–efficiency profile by outperforming WRN-28-4, ResNet-32 $\times$ 4, ConvNeXt-Tiny-CIFAR, and ViT-T/2 while utilizing significantly fewer MACs. Beyond the raw performance metrics, the controlled ablations reveal the crucial role of geometric inductive biases in representation learning. Replacing the Householder transformation with windowed MHSA or head-wise linear mixing reduces accuracy to 82.16% and 82.95% respectively (a performance drop of 1.94% and 1.15%), despite both alternatives possessing strictly greater expressive capacity (reflected in their higher parameter and MAC counts). This implies that in low-resolution and limited-data regimes,

Table 3: Tiny ImageNet comparison under 300-epoch training from scratch. Input resolution is shown in parentheses.

Method (input)	Params (M)	MACs (G)	Top-1 (%)
<i>Convolutional networks</i>			
ResNet-18-Small (64)	11.2714	2.2218	66.20
MobileNetV2-1.4 (64)	4.6744	0.0477	54.12
ConvNeXtV2-Femto (64)	4.9242	1.0137	67.22
<i>Vision Transformers and hybrid models</i>			
DeiT-Tiny/4 (64)	5.4363	1.6710	54.82
ViT-Tiny (224)	5.5561	1.2535	59.00
ViT-Tiny + DWConv (224)	5.5838	1.2576	64.10
CaiT-XXS12 (224)	6.4503	1.2763	59.17
CaiT-XXS12 + DWConv (224)	6.4345	1.2804	68.90
MobileViT-S (64)	5.1247	0.1079	55.73
<i>Householder models and matched ablations</i>			
HOVNet (Ours) (64)	5.6610	1.1540	65.03
HOVNET-MHSA (Ours, 64)	6.6677	1.5032	63.00
HOVNET-HeadLinear (Ours, 64)	5.9677	1.1902	64.32
HOVNET-Hybrid (Ours, 224)	5.8463	1.0816	69.55

the unconstrained accessible dimensions of dense linear mixers or self-attention mechanisms render them prone to overfitting or inefficient local minima. By contrast, the Householder operator introduces a strict, scale-invariant orthogonal constraint. This architecture-level restriction on the accessible Lie dimension effectively regularizes the hypothesis space during channel mixing, demonstrating that a rigorously bounded mathematical structure provides a more effective inductive bias than raw representational flexibility.

6.4 Tiny ImageNet Classification

Table 3 compares HOVNET with convolutional networks and compact vision Transformers (He et al., 2016; Sandler et al., 2018; Woo et al., 2023; Dosovitskiy et al., 2021; Touvron et al., 2021a;b; Mehta & Rastegari, 2022; Zhang et al., 2025). All models are trained from scratch for 300 epochs. Our models use AdamW with an initial learning rate of 5×10^{-4} , 20 warmup epochs, cosine decay, and an effective batch size of 512.

Data processing. Models within each input-resolution setting use color jitter, RandAugment, Random Erasing, Mixup, and CutMix with the same hyperparameters.

The base HOVNET retains the three-stage multi-head Householder backbone used on CIFAR-100, with only the stem stride and classifier adapted to the 64×64 input and 200 classes.

To evaluate the architectural modularity of our proposed operator, we introduce an additional variant, HOVNET-Hybrid, designed for the higher-resolution 224×224 setting. We construct this model by adopting the strong CaiT-XXS12 + DWConv architecture as a baseline. We retain its convolutional stem and the final Class-Attention (CA) stages, but entirely strip out its repeated patch-level multi-head self-attention (MHSA) modules. In their place, we insert our Householder blocks to handle all patch-level channel mixing.

At the native 64×64 resolution, the entirely attention-free HOVNET achieves 65.03% top-1 accuracy. This result is 1.17 and 2.19 percentage points below ResNet-18-Small and ConvNeXtV2-Femto, respectively, while outperforming MobileNetV2-1.4 and the listed 64×64 vision-Transformer baselines.

More critically, the performance of the HOVNET-Hybrid model at 224×224 resolution highlights the robust routing capabilities of the Householder constraint. By substituting standard patch-level self-attention with Householder blocks, the Hybrid model sheds 9.1% of its parameters and 15.5% of its MACs compared to

its CaiT baseline, yet improves global accuracy to 69.55%. From a network design perspective, this proves that strict orthogonal operations enforce a non-redundant feature transformation that effectively prevents dimensional collapse. Rather than relying on data-heavy attention mechanisms to separate representations, the rigid operator-level constraint inherently decouples channel features. This enables the Householder operator to serve as a highly efficient and substitutable information router, maintaining strong representational integrity whether deployed as an independent backbone or seamlessly integrated as a localized module within a broader global token architecture.

7 Conclusion

This paper bridges the differential geometry of training dynamics and practical neural network design by formalizing the accessible Lie dimension. While our theoretical analysis reveals that standard analytic architectures generically exhaust their local parameter directions, we demonstrate that this local redundancy can be actively controlled. We establish two constructive reduction strategies—confining optimization to symmetry-fixed invariant subspaces through Pairing and embedding common first integrals through HOV-NET—which serve as principled blueprints for injecting strict geometric inductive biases. Our empirical results show that deliberately restricting the accessible Lie dimension does not stifle capacity, but rather regularizes the hypothesis space to yield highly competitive accuracy-efficiency trade-offs. Ultimately, this work provides a broader theoretical framework for understanding how architectural choices dictate optimization dynamics, paving the way for more mathematically rigorous and compact deep learning systems.

References

- Yoshua Bengio, Aaron Courville, and Pascal Vincent. Representation learning: A review and new perspectives. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 35(8):1798–1828, 2013. doi: 10.1109/TPAMI.2013.50.
- Gregory Cohen, Saeed Afshar, Jonathan Tapson, and André van Schaik. EMNIST: Extending MNIST to handwritten letters. In *2017 International Joint Conference on Neural Networks (IJCNN)*, pp. 2921–2926. IEEE, 2017. doi: 10.1109/IJCNN.2017.7966217.
- Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16×16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*, 2021.
- Charles Fefferman, Sanjoy Mitter, and Hariharan Narayanan. Testing the manifold hypothesis. *Journal of the American Mathematical Society*, 29(4):983–1049, 2016. doi: 10.1090/jams/852.
- Charles Godfrey, Davis Brown, Tegan Emerson, and Henry Kvinge. On the symmetries of deep learning models and their internal representations. In *Advances in Neural Information Processing Systems*, 2022.
- Guy Gur-Ari, Daniel A. Roberts, and Ethan Dyer. Gradient descent happens in a tiny subspace. *arXiv preprint arXiv:1812.04754*, 2018.
- Ali Hassani, Steven Walton, Nikhil Shah, Abulikemu Abuduweili, Jiachen Li, and Humphrey Shi. Escaping the big data paradigm with compact transformers. *arXiv preprint arXiv:2104.05704*, 2021a.
- Ali Hassani, Steven Walton, Nikhil Shah, Abulikemu Abuduweili, Jiachen Li, and Humphrey Shi. Escaping the big data paradigm with compact transformers. *arXiv preprint arXiv:2104.05704*, 2021b.
- Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 770–778, 2016.
- Robert Hermann. On the accessibility problem in control theory. In *International Symposium on Nonlinear Differential Equations and Nonlinear Mechanics*, pp. 325–332. Academic Press, 1963.

- Alston S. Householder. Unitary triangularization of a nonsymmetric matrix. *Journal of the ACM*, 5(4): 339–342, 1958.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.
- Stephanie L. Hyland and Gunnar Rätsch. Learning unitary operators with help from $u(n)$. In *Proceedings of the Thirty-First AAAI Conference on Artificial Intelligence*, pp. 2050–2058, 2017.
- Arthur Jacot, Franck Gabriel, and Clément Hongler. Neural tangent kernel: Convergence and generalization in neural networks. In *Advances in Neural Information Processing Systems*, 2018.
- Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009.
- Ya Le and Xuan Yang. Tiny imagenet visual recognition challenge. CS231N Course Project, Stanford University, 2015.
- Chunyuan Li, Heerad Farkhoor, Rosanne Liu, and Jason Yosinski. Measuring the intrinsic dimension of objective landscapes. In *International Conference on Learning Representations*, 2018.
- Tao Li, Lei Tan, Zhehao Huang, Qinghua Tao, Yipeng Liu, and Xiaolin Huang. Low dimensional trajectory hypothesis is true: DNNs can be trained in tiny subspaces. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(3):3411–3420, 2023.
- Zhuang Liu, Hanzi Mao, Chao-Yuan Wu, Christoph Feichtenhofer, Trevor Darrell, and Saining Xie. A ConvNet for the 2020s. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 11976–11986, 2022.
- Zhiying Lu, Hongtao Xie, Chuanbin Liu, and Yongdong Zhang. Bridging the gap between vision transformers and convolutional neural networks on small datasets. In *Advances in Neural Information Processing Systems*, 2022.
- Sibylle Marcotte, Rémi Gribonval, and Gabriel Peyré. Abide by the law and follow the flow: Conservation laws for gradient flows. In *Advances in Neural Information Processing Systems*, 2023.
- Sachin Mehta and Mohammad Rastegari. MobileViT: Light-weight, general-purpose, and mobile-friendly vision transformer. In *International Conference on Learning Representations*, 2022.
- Tadashi Nagano. Linear differential systems with singularities and an application to transitive Lie algebras. *Journal of the Mathematical Society of Japan*, 18(4):398–404, 1966.
- Duy-Kien Nguyen, Mahmoud Assran, Unnat Jain, Martin R. Oswald, Cees G. M. Snoek, and Xinlei Chen. An image is worth more than 16×16 patches: Exploring transformers on individual pixels. In *International Conference on Learning Representations*, 2025.
- Mark Sandler, Andrew Howard, Menglong Zhu, Andrey Zhmoginov, and Liang-Chieh Chen. Mobilenetv2: Inverted residuals and linear bottlenecks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 4510–4520, 2018.
- Andrew M. Saxe, James L. McClelland, and Surya Ganguli. Exact solutions to the nonlinear dynamics of learning in deep linear neural networks. In *International Conference on Learning Representations*, 2014.
- Joshua B. Tenenbaum, Vin de Silva, and John C. Langford. A global geometric framework for nonlinear dimensionality reduction. *Science*, 290(5500):2319–2323, 2000. doi: 10.1126/science.290.5500.2319.
- Hugo Touvron, Matthieu Cord, Matthijs Douze, Francisco Massa, Alexandre Sablayrolles, and Hervé Jégou. Training data-efficient image transformers and distillation through attention. In *Proceedings of the 38th International Conference on Machine Learning*, Proceedings of Machine Learning Research, pp. 10347–10357. PMLR, 2021a.

- Hugo Touvron, Matthieu Cord, Alexandre Sablayrolles, Gabriel Synnaeve, and Hervé Jégou. Going deeper with image transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 32–42, 2021b.
- Wenhai Wang, Enze Xie, Xiang Li, Deng-Ping Fan, Kaitao Song, Ding Liang, Tong Lu, Ping Luo, and Ling Shao. Pyramid vision transformer: A versatile backbone for dense prediction without convolutions. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 568–578, 2021.
- Frank W. Warner. *Foundations of Differentiable Manifolds and Lie Groups*. Springer, New York, 1983.
- Sanghyun Woo, Shoubhik Debnath, Ronghang Hu, Xinlei Chen, Zhuang Liu, In So Kweon, and Saining Xie. ConvNeXt V2: Co-designing and scaling ConvNets with masked autoencoders. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 16133–16142, 2023.
- Haiping Wu, Bin Xiao, Noel Codella, Mengchen Liu, Xiyang Dai, Lu Yuan, and Lei Zhang. CvT: Introducing convolutions to vision transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 22–31, 2021.
- Sergey Zagoruyko and Nikos Komodakis. Wide residual networks. In *Proceedings of the British Machine Vision Conference*, pp. 87.1–87.12. BMVA Press, 2016. doi: 10.5244/C.30.87.
- Bowei Zhang and Yi Zhang. MSCViT: A small-size ViT architecture with multi-scale self-attention mechanism for tiny datasets. *Neural Networks*, 188:107499, 2025. doi: 10.1016/j.neunet.2025.107499.
- Tianxiao Zhang, Wenju Xu, Bo Luo, and Guanghui Wang. Depth-wise convolutions in vision transformers for efficient training on small datasets. *Neurocomputing*, 617:128998, 2025. doi: 10.1016/j.neucom.2024.128998.
- Zizhao Zhang, Han Zhang, Long Zhao, Ting Chen, Sercan Ö. Arik, and Tomas Pfister. Nested hierarchical transformer: Towards accurate, data-efficient and interpretable visual understanding. In *Proceedings of the AAAI Conference on Artificial Intelligence*, number 3, pp. 3417–3425, 2022. doi: 10.1609/aaai.v36i3.20252.
- Jiajie Zhao, Tao Luo, and Yaoyu Zhang. Architecture induces structural invariant manifolds of neural network training dynamics. *Mathematical Models and Methods in Applied Sciences*, 2026. doi: 10.1142/S0218202526420078. Advance online publication.

A Proofs of the Main Results

Let $f_\theta : \mathcal{X} \rightarrow \mathbb{R}^c$, with $\theta \in \Theta \subseteq \mathbb{R}^M$, and define

$$X_{x,a}(\theta) = \nabla_\theta \langle a, f_\theta(x) \rangle, \quad (12)$$

$$\mathcal{W} = \{X_{x,a} : x \in \mathcal{X}, a \in \mathbb{R}^c\}. \quad (13)$$

The accessible distribution and accessible Lie dimension are

$$\mathcal{D}(\theta) = \text{span}\{Y(\theta) : Y \in \text{Lie}(\mathcal{W})\}, \quad d_{\text{Lie}}(\theta) = \dim \mathcal{D}(\theta).$$

A.1 Proof of Theorem 1

Theorem 2 (Generic maximality). *Let $\Theta \subseteq \mathbb{R}^M$ be connected and open, and assume that every vector field in $\mathcal{W}$ is real analytic on Θ . Define*

$$R = \max_{\theta \in \Theta} d_{\text{Lie}}(\theta), \quad \Omega_R = \{\theta \in \Theta : d_{\text{Lie}}(\theta) = R\}.$$

Then Ω_R is open and $\Theta \setminus \Omega_R$ has Lebesgue measure zero. Moreover, if μ is a probability measure on Θ that is absolutely continuous with respect to Lebesgue measure and $\theta_0 \sim \mu$, then

$$\Pr_{\theta_0 \sim \mu}(\theta_0 \in \Omega_R) = \mu(\Omega_R) = 1.$$

Proof. Because d_{Lie} takes values in $\{0, \dots, M\}$, its maximum is attained. Choose $\theta_\star \in \Theta$ such that $d_{\text{Lie}}(\theta_\star) = R$. There exist $Y_1, \dots, Y_R \in \text{Lie}(\mathcal{W})$ whose values at $\theta_\star$ are linearly independent.

Every Y_j is analytic because finite Lie brackets preserve analyticity. Therefore, some $R \times R$ coordinate minor

$$\Delta(\theta) = \det(Y_j^{i_k}(\theta))_{k,j=1}^R$$

is analytic and satisfies $\Delta(\theta_\star) \neq 0$. Hence Δ is not identically zero. Whenever $\Delta(\theta) \neq 0$, the vectors $Y_1(\theta), \dots, Y_R(\theta)$ remain linearly independent. Thus $d_{\text{Lie}}(\theta) \geq R$, and maximality of R implies $\theta \in \Omega_R$. Consequently,

$$\Theta \setminus \Omega_R \subseteq \Delta^{-1}(0).$$

The zero set of a nonzero real-analytic function on a connected open set has Lebesgue measure zero.

To prove openness, fix $\theta \in \Omega_R$. Repeating the preceding construction at θ gives an $R \times R$ analytic minor that is nonzero at θ . It remains nonzero on a neighborhood of θ , and that neighborhood is contained in Ω_R . Finally, absolute continuity of μ gives $\mu(\Theta \setminus \Omega_R) = 0$, which proves the probability statement. $\square$

Finite-horizon consequence. If every marginal distribution of θ_t , $t = 0, \dots, T$, is absolutely continuous with respect to Lebesgue measure, then the theorem above and the union bound imply

$$\Pr(\theta_t \in \Omega_R \text{ for all } t = 0, \dots, T) = 1.$$

No independence assumption between the iterates is required.

A.2 Proof of Proposition 1

Proposition 3 (First-integral restriction). *If C is a regular common first integral of $\mathcal{W}$, then*

$$dC_\theta(Y(\theta)) = 0, \quad \forall \theta \in \Theta, \quad Y \in \text{Lie}(\mathcal{W}).$$

Consequently,

$$\mathcal{D}(\theta) \subseteq \ker dC_\theta, \quad d_{\text{Lie}}(\theta) \leq M - k, \quad \rho_{\text{Lie}}(\theta) \leq 1 - \frac{k}{M}.$$

Proof. Write $C = (C_1, \dots, C_k)$. The common-first-integral assumption is equivalent to

$$X(C_\ell) = 0 \quad \text{for every } X \in \mathcal{W} \text{ and } \ell = 1, \dots, k.$$

If two vector fields X and Y annihilate C_ℓ , then

$$[X, Y](C_\ell) = X(Y(C_\ell)) - Y(X(C_\ell)) = 0.$$

Closure under linear combinations and induction over bracket depth show that every field in $\text{Lie}(\mathcal{W})$ annihilates every component of C . Hence $\mathcal{D}(\theta) \subseteq \ker dC_\theta$. Since C is regular, $\text{rank } dC_\theta = k$, and rank-nullity gives

$$\dim \ker dC_\theta = M - k.$$

The stated dimension and ratio bounds follow. $\square$

Householder specialization

For $v \in \mathbb{R}^d \setminus \{0\}$, let

$$H(v) = I_d - 2 \frac{vv^\top}{v^\top v}.$$

Since $H(\alpha v) = H(v)$ for every $\alpha \neq 0$, any scalar probe whose dependence on v is only through $H(v)$ can be written as $g(v) = \tilde{g}(H(v))$ and satisfies

$$0 = \left. \frac{d}{d\alpha} g(\alpha v) \right|_{\alpha=1} = \langle \nabla_v g(v), v \rangle.$$

Therefore

$$C(v) = \frac{1}{2} \|v\|_2^2$$

is a common first integral.

Suppose that N_{refl} nonzero reflector vectors occupy disjoint parameter blocks and that the model depends on each such vector only through its Householder reflector. The map collecting their squared norms has differential rank N_{refl} . The first-integral proposition above therefore yields

$$d_{\text{Lie}}(\theta) \leq M - N_{\text{refl}}, \quad \rho_{\text{Lie}}(\theta) \leq 1 - \frac{N_{\text{refl}}}{M}.$$

This is the architecture-level certificate used for the Householder parameterization.

Bilinear specialization

For $\phi(Q, K) = QK^\top$, with $Q \in \mathbb{R}^{m \times r}$ and $K \in \mathbb{R}^{n \times r}$, define

$$C(Q, K) = \text{vech}(Q^\top Q - K^\top K),$$

where vech vectorizes the independent entries of a symmetric matrix. If a scalar probe depends on Q and K only through $QK^\top$, then direct differentiation shows that $C(Q, K)$ is constant along every induced gradient field. Wherever $\text{rank } dC_{(Q, K)} = k$, the first-integral proposition above gives

$$d_{\text{Lie}}(Q, K) \leq M - k.$$

A.3 Proof of Proposition 2

Let $\rho : \Gamma \rightarrow O(M)$ be an orthogonal representation satisfying $\rho(\gamma)\Theta = \Theta$, and define

$$F_\Gamma = \{v \in \mathbb{R}^M : \rho(\gamma)v = v, \forall \gamma \in \Gamma\}, \quad \Theta^\Gamma = \Theta \cap F_\Gamma.$$

Proposition 4 (Symmetry restriction and preservation). *Suppose that the model is Γ -invariant. Then*

$$\mathcal{D}(\theta) \subseteq F_\Gamma, \quad d_{\text{Lie}}(\theta) \leq \dim F_\Gamma$$

for every $\theta \in \Theta^\Gamma$. Consequently,

$$\rho_{\text{Lie}}(\theta) \leq \frac{\dim F_\Gamma}{M} < 1$$

on Θ^Γ whenever $F_\Gamma \subsetneq \mathbb{R}^M$.

Moreover, if $\theta_0 \in \Theta^\Gamma$ and every update $\Delta_t \in F_\Gamma$, then every iterate of $\theta_{t+1} = \theta_t + \Delta_t$ remains in Θ^Γ . Hence, the same dimension bounds hold throughout training.

Remark 3. *The update condition in the symmetry proposition above is automatically satisfied by standard gradient-based optimizers under the corresponding symmetry assumptions.*

Proof. Fix $x \in \mathcal{X}$ and $a \in \mathbb{R}^c$, and write

$$g(\theta) = \langle a, f_\theta(x) \rangle.$$

Model invariance gives $g \circ \rho(\gamma) = g$. The chain rule and orthogonality of $\rho(\gamma)$ imply

$$\nabla g(\rho(\gamma)\theta) = \rho(\gamma)\nabla g(\theta).$$

Thus every generator in $\mathcal{W}$ is Γ -equivariant.

If X and Y are Γ -equivariant, differentiating $X(\rho(\gamma)\theta) = \rho(\gamma)X(\theta)$ gives

$$DX(\rho(\gamma)\theta)\rho(\gamma) = \rho(\gamma)DX(\theta),$$

and the same identity holds for Y . Substitution into

$$[X, Y] = DY X - DX Y$$

shows that $[X, Y]$ is also equivariant. Induction over bracket depth therefore gives

$$Z(\rho(\gamma)\theta) = \rho(\gamma)Z(\theta) \quad \text{for every } Z \in \text{Lie}(\mathcal{W}).$$

At $\theta \in \Theta^\Gamma$, this becomes

$$Z(\theta) = \rho(\gamma)Z(\theta) \quad \forall \gamma \in \Gamma,$$

so $Z(\theta) \in F_\Gamma$. Taking spans proves $\mathcal{D}(\theta) \subseteq F_\Gamma$, and the dimension and ratio bounds follow.

For the discrete statement, F_Γ is a linear subspace. Therefore, if $\theta_t, \Delta_t \in F_\Gamma$, then $\theta_{t+1} = \theta_t + \Delta_t \in F_\Gamma$. Induction from $\theta_0 \in F_\Gamma$ proves preservation at every iteration. $\square$

Pairing as a permutation-fixed construction

A single pair is generated by a transposition matrix P , and several disjoint pairs generate a product of copies of $\mathbb{Z}_2$. For two consecutive affine layers,

$$z = \phi(Wx + b), \quad y = Uz + c,$$

the parameter action is

$$(W, b, U, c) \mapsto (PW, Pb, UP^\top, c).$$

Because ϕ is coordinatewise, $\phi(Pu) = P\phi(u)$, and therefore

$$UP^\top \phi(P(Wx + b)) + c = U\phi(Wx + b) + c.$$

The network function is invariant for every parameter value. The fixed-point equations are

$$W_{j\cdot} = W_{i\cdot}, \quad b_j = b_i, \quad U_{\cdot j} = U_{\cdot i},$$

which is neuron Pairing. The same calculation applies to a Transformer feed-forward network with $W = W_1$ and $U = W_2$.

For multihead attention, let P_H permute heads and let $\Pi = P_H \otimes I_d$ permute the concatenated head coordinates. Permuting the Q , K , and V parameter blocks sends the concatenated head output $h(x)$ to $\Pi h(x)$. Simultaneously replacing the output projection by $W_O \Pi^\top$ leaves the output unchanged:

$$W_O \Pi^\top \Pi h(x) = W_O h(x).$$

The fixed-point equations equate the paired $Q/K/V$ blocks and the corresponding column blocks of W_O . Hence MLP-neuron, Transformer-FFN-unit, and attention-head Pairing all satisfy the invariance hypothesis of the symmetry proposition above.

Lemma 1 (Permutation equivariance of standard optimizers). *Let P be a coordinate-permutation matrix. Suppose the stochastic gradient satisfies*

$$g_t(P\theta) = Pg_t(\theta),$$

paired coordinates use identical optimizer hyperparameters, and every optimizer state is initialized in the fixed subspace of P . Then SGD, momentum SGD, Adam, and AdamW preserve the fixed subspace. Global-norm gradient clipping also preserves it.

Proof. For SGD, the update

$$\theta^+ = \theta - \eta g_t(\theta)$$

commutes with P . Momentum uses

$$m^+ = \beta m + (1 - \beta)g_t(\theta), \quad \theta^+ = \theta - \eta m^+,$$

so $(P\theta, Pm)$ is mapped to $(P\theta^+, Pm^+)$.

Adam additionally maintains

$$v^+ = \beta_2 v + (1 - \beta_2) g_t(\theta) \odot g_t(\theta).$$

Coordinate permutations commute with elementwise products, square roots, division, and scalar bias corrections. Hence the Adam update is equivariant. AdamW adds a term proportional to $-\theta$, which also commutes with P . Finally, $\|Pg\|_2 = \|g\|_2$, so multiplication by a global clipping factor is equivariant. Zero first- and second-moment states are fixed by every permutation, and induction completes the proof. $\square$

For empirical-risk training, every per-example loss inherits the parameter symmetry, so every minibatch gradient obeys the required equivariance. Exact pathwise preservation additionally requires stochastic coordinatewise operations, such as dropout, to use a shared or permutation-equivariant mask. With independently sampled masks, the deterministic pathwise conclusion of Proposition 2 need not apply. This does not by itself invoke the finite-horizon generic-maximality consequence: that consequence requires the parameter marginal at each iterate to be absolutely continuous with respect to ambient Lebesgue measure, whereas Bernoulli dropout produces discrete, structured noise and need not establish such absolute continuity. The Pairing experiment description below states the behavior of the stochastic operators used in our actual experiments.